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Dynamic Resume Builder: A Frontend-Based Web Application for Real-Time Resume Creation and PDF Export

A Software Engineering Project Submitted
By

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Course: Software Engineering | Semester: Spring 2026

1. Project Title

Dynamic Resume Builder: A Frontend-Based Web Application for Real-Time Resume Creation and PDF Export

2. Introduction

2.1 Overview oF the Domain

The job recruitment landscape has become increasingly competitive, with employers receiving hundreds of applications for a single position. As a result, structured, well-formatted resumes play a critical role in the initial screening process. Many organizations now use Applicant Tracking Systems (ATS) to filter candidates based on formatting consistency, keyword presence, and structural clarity. Traditional resume creation methods using word processors often lead to inconsistent layouts, formatting errors, and compatibility issues during PDF conversion. Users frequently struggle with alignment, spacing, font consistency, and section organization. Modern web technologies such as React and Tailwind CSS enable the development of dynamic, responsive interfaces that simplify structured content creation while maintaining professional formatting standards. This project focuses on developing a frontend-based web application that allows users to create, preview, and export professionally formatted resumes using dynamic forms and client-side PDF generation.

2.2 Motivation of the Project

Building a dynamic resume builder application provides an ideal opportunity to apply core Software Engineering principles — requirements engineering, component-based system design, implementation, testing, and deployment — within a practical academic context. Many individuals rely on traditional word processors that require manual formatting, leading to inconsistent layouts and inefficient editing workflows. Existing online resume tools often include subscription fees, limited template customization, and restricted export functionality. This project demonstrates that a structured, frontend-based application can deliver professional resume creation capabilities within an academic semester, while also serving as a portfolio-ready project showcasing modern React-based development and client-side document generation proficiency.

3. Background to the Problem

3.1 Current System Overview

Most job seekers currently create resumes using traditional tools such as Microsoft Word or Google Docs, manually adjusting formatting, alignment, spacing, and typography. Others rely on free online resume builders that provide limited design options or require account registration and paid upgrades for full functionality. In many cases, exported documents may suffer from layout inconsistencies, font mismatches, or formatting errors during PDF conversion. These approaches lack structured content management, standardized template rendering, and real-time preview capabilities.

3.2 Limitations of the Existing System

Key limitations include: (a) inconsistent formatting due to manual layout adjustments; (b) lack of centralized, structured input sections for resume components; (c) limited customization options in free online tools; (d) unreliable PDF export formatting across different devices or browsers; (e) absence of real-time preview while editing content; and (f) no persistent local data management for structured resume updates across sessions.

3.3 Need For the Proposed Solution

A dedicated resume builder application addresses these limitations by providing structured input sections for personal details, education, experience, and skills; a real-time split-screen preview system; multiple standardized templates; client-side PDF generation; and local data persistence for seamless editing. The system ensures consistent formatting, improved usability, and professional output — all within a lightweight, frontend-heavy architecture that requires no recurring subscription fees or external dependencies.

4. Problem Statement

Students and job seekers lack a structured, standardized platform for creating professionally formatted resumes efficiently. Traditional word processors require manual formatting and layout management, increasing the risk of inconsistencies and design errors. Without a centralized system, users struggle with maintaining uniform section alignment, previewing real-time updates, and ensuring reliable PDF export formatting. Existing online resume tools often impose feature limitations or paid upgrades, reducing accessibility and customization flexibility. A dedicated, frontend-based resume builder application is needed to provide dynamic content management, standardized template rendering, real-time preview functionality, and consistent client-side PDF export while maintaining simplicity and full user control.

5. Objectives of the Project

1. Develop a responsive resume input system with structured sections for personal information, education, experience, and skills.
2. Implement real-time split-screen preview with dynamic template rendering.
3. Integrate client-side PDF generation with consistent formatting and layout preservation.
4. Design multiple resume templates with standardized styling and formatting rules.
5. Implement centralized state management using Context API for structured data handling.
6. Enable persistent data storage using LocalStorage to retain resume information across sessions.

6. Proposed Solution Overview

The proposed system is a frontend-heavy web application built using React and Tailwind CSS, following a modular component-based architecture. The application will utilize a centralized state management approach through the Context API to manage resume data dynamically across multiple components. The system consists of a split-screen interface where users enter resume information through structured input forms on one side, while a real-time rendered preview is displayed on the other. Template rendering logic dynamically maps structured data into standardized resume layouts. Client-side PDF generation is implemented using html2canvas and jsPDF to ensure consistent export formatting. Data persistence is handled through LocalStorage, allowing users to retain resume information across browser sessions without requiring backend infrastructure.

Technology Stack

Component	Technology	Rationale
Frontend	ReacL.js	Component-based UI with dynamic rendering
Styling	Tailwind CSS	Utility-first responsive design framework
State Management	Context API	Centralized state control across components
PDF Generation	html2canvas + jsPDF	Client-side document rendering and export
Data Storage	LocalStorage	Persistent data retention without backend
Deployment	Vercel / Netlify	Simple static frontend deployment

7. Target Users

7.1 Primary Users

Students: Create structured resumes for internships and entry-level positions.
Job Seekers: Develop professionally formatted resumes for job applications.
Professionals: Update and maintain standardized resumes efficiently.Secondary Users.

7.2 Secondary Users

Career Counselors: Assist students in resume preparation and formatting.
Academic Institutions: Use as a demonstration tool for career development workshops.

8. Scope of the Project

8.1 In-Scope

Structured resume input forms (Personal Information, Education, Experience, Skills), Real-time split-screen preview with dynamic template rendering, Multiple standardized resume templates, Client-side PDF export with layout consistency, LocalStorage-based data persistence, Responsive UI design for desktop and tablet devices, Frontend-only deployment on cloud hosting platforms.

8.2 Out Scope

Backend database integration, User authentication or account management system, Cloud-based resume storage, AI-based resume analysis or content generation, Multi-language support, Mobile native application development.

9. Expected Impact and Benefits

Operational: Streamlined resume creation process with reduced formatting errors and structured content management.
User Experience: Real-time preview enhances usability and reduces editing friction. Consistent PDF export ensures professional output.
Academic: Demonstrates application of core Software Engineering principles including modular architecture, centralized state management, UI responsiveness, and client-side processing.
Portfolio Value: Serves as a practical demonstration of React-based frontend development and structured application design.

10. Conclusion

This proposal presents a dynamic, frontend-based Resume Builder application designed to address common formatting challenges associated with traditional resume creation tools. By integrating structured input forms, real-time preview rendering, multiple templates, and client-side PDF export, the system ensures consistent and professional resume output. The scope is well-defined for completion within a single academic semester, and the chosen technologies are aligned with modern frontend development practices. Upon completion, the project will serve as both a functional productivity tool and a comprehensive demonstration of applied Software Engineering concepts.